A Appendix

Example 1.
Consider a game with a set of two agents $N = \{1, 2\}$ and a set of two projects $M = \{a, b\}$, where $r(a) = r(b) = 1$. Player 1 is happy to participate in any project as long as she is alone, and player 2 always wants to participate in a project with player 1. Formally, player 1 strictly prefers working alone with $p_1(\{1\}) = 1$, while player 2 prefers to collaborate with player 1, i.e., $p_2(\{1, 2\}) = 1$.

Initially, consider the strategy profile $\mathbf{s}^0 = (s_1, s_2) = (a, b)$, where player 1 selects project a and player 2 selects project b. The corresponding utilities are $u_1(\mathbf{s}^0) = 1$ and $u_2(\mathbf{s}^0) = 0$, since player 2 prefers to collaborate with player 1. To improve their utility, player 2 unilaterally deviates to project a, forming the new strategy profile $\mathbf{s}^1 = (a, a)$. The utilities update to $u_1(\mathbf{s}^1) = 0$ and $u_2(\mathbf{s}^1) = 0.5$. Player 1, experiencing a reduction in utility, now deviates to project b, resulting $\mathbf{s}^2 = (b, a)$, where $u_1(\mathbf{s}^2) = 1$ and $u_2(\mathbf{s}^2) = 0$. The process repeats, with player 2 subsequently deviating to b, leading to $\mathbf{s}^3 = (b, b)$, where $u_1(\mathbf{s}^3) = 0$ and $u_2(\mathbf{s}^3) = 0.5$, prompting player 1 to revert to project a, thus restarting the cycle (see Table 1).

Players	$\mathbf{s}^0$	$\mathbf{s}^1$	$\mathbf{s}^2$	$\mathbf{s}^3$
1	a	a	b	b
2	b	a	a	b

Table 1: The sequence of improving unilateral deviations

Example 2.
Consider a hedonic project game with agents $N = \{1, 2, 3\}$ and projects $M = \{a, b\}$, where $r(a) = r(b) = 10$. Each agent assigns positive preference to exactly one two-player coalition and zero to all others:

$$p_1(\{1, 2\}) = 1, \quad p_2(\{2, 3\}) = 1, \quad p_3(\{1, 3\}) = 1.$$

Consider the strategy profile $\mathbf{s} = (a, a, a)$. Since no agent assigns positive preference to the grand coalition, we have $u_i(\mathbf{s}) = 0$ for all $i \in N$. Moreover, any unilateral deviation yields a singleton coalition and hence zero utility. Thus, $\mathbf{s} \in \mathsf{NS}(G)$.

Now consider $\mathbf{s}^\star = (a, a, b)$. Then $C_a(\mathbf{s}^\star) = \{1, 2\}$ and $\mathrm{SW}(\mathbf{s}^\star) = p_1(\{1, 2\}) \cdot \frac{10}{2} + p_2(\{1, 2\}) \cdot \frac{10}{2} = 5$. Hence $\mathsf{OPT}(G) = 5$, while $\mathrm{SW}(\mathbf{s}) = 0$, implying $\mathsf{PoA}_{\mathrm{NS}}(G) = \infty$.

Example 3.
Consider a hedonic project game with agents $N = \{1, 2, 3\}$ and projects $M = \{a, b\}$, where $r(a) = r(b) = 1$. For each agent $i \in N$ and $\ell \neq i$, preferences are:

$$p_i(\{i\}) = 0.5, \quad p_i(\{i, \ell\}) = 0.182, \quad p_i(N) = 0.136.$$

Any outcome in which all agents select the same project is not Nash stable, since any agent can deviate to the empty project and strictly increase her utility.

Now consider an outcome $\mathbf{s}$ in which two agents form a pair on one project and the remaining agent is alone on the other project. Each agent in the pair obtains

$$u_i(\mathbf{s}) = 0.182 \cdot \frac{1}{2} = 0.091$$

If such an agent deviates to the singleton project, she again forms a coalition of size two and remains indifferent: $u_i(\mathbf{s}') = 0.091$. Hence, $\mathbf{s}$ is Nash stable.

However, the agent left behind strictly benefits from the deviation. Her utility increases from $u_\ell(\mathbf{s}) = 0.091$ to

$$u_\ell(\mathbf{s}') = 0.5$$

Thus, the deviation strictly improves the utility of a non-deviating agent, violating the leave-stability condition. Consequently, no leaving stable outcome exists.

Example 4.
Consider a hedonic project game with agents $N = \{1, 2, 3\}$ and projects $M = \{a, b\}$. Both projects have identical rewards $r(a) = r(b) = 1$. Each agent $i \in N$ has per-capita non-decreasing preferences. Agent 1's preferences are:

$$\hat{p}_1(\{1\}) = \hat{p}_1(\{1, \ell\}) = \hat{p}_1(N) = 0.125, \quad \ell \in \{2, 3\}$$

Each agent $\ell \in \{2, 3\}$ has strictly non-decreasing per-capita preferences:

$$p_\ell(\{\ell\}) = 0.11, \quad p_\ell(\{\ell, 1\}) = 0.22,$$
$$p_\ell(\{2, 3\}) = 0.22, \quad p_\ell(N) = 0.44.$$

Consider the outcome $\mathbf{s} = (b, a, a)$ in which agent 1 is assigned to project b, and agents 2 and 3 are assigned to project a, i.e., $C_a(\mathbf{s}) = \{2, 3\}$ and $C_b(\mathbf{s}) = \{1\}$. The resulting utilities are

$$u_1(\mathbf{s}) = 0.125, \quad u_2(\mathbf{s}) = u_3(\mathbf{s}) = 0.22$$

Agent 1 is indifferent between staying at b and deviating to a, since

$$u_1(a, \mathbf{s}_{-1}) = \hat{p}_1(N) \cdot r(a) = 0.125 = u_1(\mathbf{s})$$

Agents 2 and 3 are indifferent between staying on project a and deviating to project b, since deviating to b would form coalition $\{1, \ell\}$ with agent 1, yielding utility $p_\ell(\{1, \ell\}) \cdot r(b) = 0.22 = u_\ell(\mathbf{s})$ (no strict improvement). Hence, the outcome $\mathbf{s}$ is NS.

Consider agent 1 joining project a. For each incumbent agent $\ell \in \{2, 3\}$, utility strictly increases:

$$p_\ell(N) \cdot r(a) = 0.44 > p_\ell(\{2, 3\}) \cdot r(a) = 0.22$$

Since $u_\ell(\mathbf{s}) < u_\ell(a, \mathbf{s}_{-1})$, the joining stability condition $u_\ell(\mathbf{s}) \geq u_\ell(a, \mathbf{s}_{-1})$ is violated. Thus, the outcome $\mathbf{s}$ is not JS.

Proof of Observation 1
Proof. Given a hedonic project game $G \in \mathscr{G}$, consider a strategy profile $\mathbf{s}$ where all agents choose the same project $j \in M$, i.e., $s_i = j$ for all $i \in N$. Thus, $C_j(\mathbf{s}) = N$, and $C_{j'}(\mathbf{s}) = \emptyset$ for all $j' \neq j$.

For Nash stability Since $w_i(\ell) \geq 0$ and due to normalization $\sum_{C \in \mathcal{N}_i} p_i(C) = 1$ for all $i \in N$, we have $p_i(N) > 0$, and by definition we also have $r(j) > 0$ for all $j \in M$. Thus, we have $u_i(\mathbf{s}) > 0$, for all $i \in N$.

If an agent i deviates to a different project $j' \neq j$, and form a singleton coalition $\{i\}$, her utility becomes $u_i(j', \mathbf{s}_{-i}) = 0$, since $w_i(i) = 0$ by definition. Since $u_i(\mathbf{s}) > 0$, it follows that $u_i(\mathbf{s}) > u_i(j', \mathbf{s}_{-i})$ for all $j' \in M$, satisfying the Nash stability condition.

For joining stability Since deviations to $j' \neq j$ involve empty coalitions ($C_{j'}(\mathbf{s}) = \emptyset$), there are no agents ℓ to violate joining stability. For $j' = j$, the condition holds trivially. Thus, the strategy profile $\mathbf{s}$ where all agents choose the same project (the grand coalition) is a joining stable outcome.

For leaving stability Suppose agent i leaves project j. The remaining agents in $C_j(\mathbf{s}) \setminus \{i\}$ may benefit (since $|C_j(\mathbf{s})|$ decreases). However, the agent i strictly loses utility, as argued in the Nash stability case. Thus, any strategy profile $\mathbf{s}$ that forms the grand coalition resists profitable deviations under leaving stability.

Thus, any strategy profile $\mathbf{s}$ that forms the grand coalition is Nash, joining, and leaving stable outcome. $\square$

Proof of Observation 2

Proof. To establish $\mathsf{NS}(G) = \mathsf{JS}(G)$, we prove the following inclusions: (1) $\mathsf{JS}(G) \subseteq \mathsf{NS}(G)$, and (2) $\mathsf{NS}(G) \subseteq \mathsf{JS}(G)$. Since $\mathsf{JS}(G) \subseteq \mathsf{NS}(G)$ holds by definition (as both include the deviator's condition), we focus on proving $\mathsf{NS}(G) \subseteq \mathsf{JS}(G)$.

Suppose $\mathbf{s} \in \mathsf{NS}(G)$. Consider a deviation by agent i from s_i to $j \neq s_i$, forming $\mathbf{s}' = (j, \mathbf{s}_{-i})$. When i deviates to j, the coalition at j becomes $C_j(\mathbf{s}') = C_j(\mathbf{s}) \cup \{i\}$, so $|C_j(\mathbf{s})| + 1$. Consider two cases for a player $\ell \in N$:

- **Case 1:** $\ell = i$ **(the deviator)**. Since $\mathbf{s} \in \mathsf{NS}(G)$, $u_i(\mathbf{s}) \geq u_i(\mathbf{s}')$, which satisfies the joining stability condition for $\ell = i$.
- **Case 2:** $\ell \in C_j(\mathbf{s})$ **(agents in the target coalition)**. Before i's deviation, ℓ's utility is:

$$u_\ell(\mathbf{s}) = p_\ell(C_j(\mathbf{s})) \cdot \frac{r(j)}{|C_j(\mathbf{s})|}$$

After i switches to j, ℓ's utility becomes:

$$u_\ell(\mathbf{s}') = p_\ell(C_j(\mathbf{s}) \cup \{i\}) \cdot \frac{r(j)}{|C_j(\mathbf{s})| + 1}$$

Since $p_\ell(\cdot)$ is monotonic decreasing, $p_\ell(C_j(\mathbf{s}) \cup \{i\}) \leq p_\ell(C_j(\mathbf{s}))$. Also, $\frac{r(j)}{|C_j(\mathbf{s})|+1} < \frac{r(j)}{|C_j(\mathbf{s})|}$ (since $|C_j(\mathbf{s})| \geq 0$). Thus:

$$u_\ell(\mathbf{s}') < u_\ell(\mathbf{s})$$

This holds for all $\ell \in C_j(\mathbf{s})$, satisfying the joining stability condition for the target coalition.

If $C_j(\mathbf{s}) = \emptyset$, only $\ell = i$ is covered by Nash stability. Thus, any deviation to j does not improve the utility of the deviating player i, and no member of the target coalition $C_j(\mathbf{s})$ would benefit from the joining of the player i, so $\mathbf{s} \in \mathsf{JS}(G)$. Thus, $\mathsf{NS}(G) \subseteq \mathsf{JS}(G)$. $\square$

Proof of Observation 3

Proof. Let $j^\star \in \arg\max_{j \in M} r(j)$ and let $\mathbf{s}^\star$ be the strategy profile in which every agent selects project $j^\star$, so that $C_{j^\star}(\mathbf{s}^\star) = N$ and all other projects are empty.

For any outcome $\mathbf{s}$, social welfare can be written as

$$\mathrm{SW}(\mathbf{s}) = \sum_{j \in M} r(j) \sum_{i \in C_j(\mathbf{s})} p_i(C_j(\mathbf{s}))$$

By per-capita non-decreasing preferences, for every agent i and every coalition $C \in \mathcal{N}_i$,

$$p_i(C) \leq p_i(N)$$

Hence, for any outcome $\mathbf{s}$,

$$\begin{aligned} \mathrm{SW}(\mathbf{s}) &\leq \sum_{j \in M} r(j) \sum_{i \in C_j(\mathbf{s})} p_i(N) \\ &\leq \sum_{i \in N} p_i(N)\, r(j^\star) = \mathrm{SW}(\mathbf{s}^\star) \end{aligned}$$

Thus, $\mathbf{s}^\star$ is socially optimal allocation.

Consider any agent i and a deviation from $j^\star$ to some project $j \neq j^\star$. After the deviation, agent i forms a singleton coalition on j, obtaining utility

$$u_i(j, \mathbf{s}^\star_{-i}) = p_i(\{i\})\, r(j)$$

Under $\mathbf{s}^\star$, agent i's utility is

$$u_i(\mathbf{s}^\star) = p_i(N)\, r(j^\star)$$

By per-capita non-decreasing preferences, $p_i(\{i\}) \leq p_i(N)$, and by the choice of $j^\star$, where $r(j^\star) \geq r(j)$ for all $j \in M$, we obtain

$$u_i(j, \mathbf{s}^\star_{-i}) \leq p_i(N)\, r(j^\star) = u_i(\mathbf{s}^\star)$$

Hence, no agent can profitably deviate, and $\mathbf{s}^\star$ is NS. $\square$

Proof of Observation 4

Proof. Let G be a hedonic project game in which all agents have per-capita strictly decreasing preferences. Let $\mathbf{s}$ be an arbitrary outcome such that there exists a project $j \in M$ with $|C_j(\mathbf{s})| \geq 2$.

Fix an agent $i \in C_j(\mathbf{s})$ and consider any unilateral deviation of i to a project $k \in M$, yielding the outcome $\mathbf{s}' = (k, \mathbf{s}_{-i})$. Let $\ell \in C_j(\mathbf{s}) \setminus \{i\}$ be any agent remaining in project j after the deviation. The utility of agent ℓ before the deviation is

$$u_\ell(\mathbf{s}) = p_\ell(C_j(\mathbf{s})) \cdot \frac{r(j)}{|C_j(\mathbf{s})|}$$

while her utility after the deviation is

$$u_\ell(\mathbf{s}') = p_\ell(C_j(\mathbf{s}) \setminus \{i\}) \cdot \frac{r(j)}{|C_j(\mathbf{s})| - 1}$$

Since preferences are per-capita strictly decreasing, for any coalition C and any strict subset $C' \subset C$,

$$\frac{p_\ell(C)}{|C|} < \frac{p_\ell(C')}{|C'|}$$

Applying this to $C = C_j(\mathbf{s})$ and $C' = C_j(\mathbf{s}) \setminus \{i\}$ and multiplying by $r(j) > 0$, we obtain

$$u_\ell(\mathbf{s}) < u_\ell(\mathbf{s}')$$

Thus, the deviation of agent i strictly increases the utility of agent ℓ, a member of i's original coalition. This violates the condition of leaving stability.

Since the argument applies to any outcome containing a coalition of size at least two, no such outcome can be leaving stable. $\square$

Proof of Lemma 1

Proof. Suppose $\mathbf{s}$ is a Nash stable outcome. We prove by contradiction that all non-empty coalitions have sizes differing by at most one. Assume there exist non-empty coalitions $C_j(\mathbf{s})$ and $C_k(\mathbf{s})$ such that $|C_j(\mathbf{s})| \geq |C_k(\mathbf{s})| + 2$. The rewards of projects are identical; without loss of generality, we consider $r(j) = 1$ for all $j \in M$. Consider an agent $i \in C_j(\mathbf{s})$. Their utility is:

$$u_i(\mathbf{s}) = \frac{p_i(C_j(\mathbf{s}))}{|C_j(\mathbf{s})|}$$

If i deviates to project k, the new coalition is $C_k(j, \mathbf{s}_{-i}) = C_k(\mathbf{s}) \cup \{i\}$, with size $|C_k(\mathbf{s})| + 1$. The utility after deviation is:

$$u_i(j, \mathbf{s}_{-i}) = \frac{p_i(C_k(\mathbf{s}) \cup \{i\})}{|C_k(\mathbf{s})| + 1}$$

Since preferences are monotonically decreasing and $|C_k(\mathbf{s})| + 1 \leq |C_j(\mathbf{s})|$, we have:

$$p_i(C_k(\mathbf{s}) \cup \{i\}) \geq p_i(C_j(\mathbf{s}))$$

Since $|C_j(\mathbf{s})| \geq |C_k(\mathbf{s})| + 2$, we have $|C_k(\mathbf{s})| + 1 \leq |C_j(\mathbf{s})| - 1$, so $\frac{1}{|C_k(\mathbf{s})|+1} \geq \frac{1}{|C_j(\mathbf{s})|-1}$. Thus:

$$u_i(j, \mathbf{s}_{-i}) \geq \frac{p_i(C_j(\mathbf{s}))}{|C_k(\mathbf{s})| + 1}$$

Compare with the current utility:

$$\frac{u_i(j, \mathbf{s}_{-i})}{u_i(\mathbf{s})} \geq \frac{\frac{p_i(C_j(\mathbf{s}))}{|C_k(\mathbf{s})|+1}}{\frac{p_i(C_j(\mathbf{s}))}{|C_j(\mathbf{s})|}} = \frac{|C_j(\mathbf{s})|}{|C_k(\mathbf{s})| + 1}$$

Since $|C_j(\mathbf{s})| \geq |C_k(\mathbf{s})| + 2$, we have:

$$\frac{|C_j(\mathbf{s})|}{|C_k(\mathbf{s})| + 1} \geq \frac{|C_k(\mathbf{s})| + 2}{|C_k(\mathbf{s})| + 1} = 1 + \frac{1}{|C_k(\mathbf{s})| + 1} > 1$$

as $|C_k(\mathbf{s})| \geq 1$. Thus, $u_i(j, \mathbf{s}_{-i}) > u_i(\mathbf{s})$, contradicting Nash stability. □

Proof of Theorem 2

Proof. We prove the claim by exhibiting a family of hedonic project games for which the price of anarchy grows arbitrarily large. Consider a game $G \in \mathscr{G}$ with two agents $N = \{1, 2\}$ and two projects $M = \{a, b\}$. Let the project rewards be $r(a) = 1$ and $r(b) = K$, where $K > 0$ is an arbitrary parameter.

Preferences are additively separable and identical for both agents. Each agent strictly prefers being in a coalition with the other agent rather than being alone. Formally, for each agent $i \in N$, define the valuation function $w_i : N \to [0, 1]$ by

$$w_i(j) = \begin{cases} 1 & \text{if } j \neq i, \\ 0 & \text{if } j = i. \end{cases}$$

Hence, for each agent i, we have $p_i(\{i\}) = 0$ and $p_i(\{1, 2\}) = 1$.

By Observation 2, any strategy profile in which all agents select the same project is NS (and also JS and LS). Consider the strategy profile $\mathbf{s} = (a, a)$, where both agents select project a. The induced coalition is $C_a(\mathbf{s}) = \{1, 2\}$. Each agent's utility is

$$u_i(\mathbf{s}) = \frac{p_i(\{1, 2\}) \cdot r(a)}{|C_a(\mathbf{s})|} = \frac{1}{2}$$

Thus, the social welfare of $\mathbf{s}$ is $\mathrm{SW}(\mathbf{s}) = 1$.

Consider the strategy profile $\mathbf{s}^\star = (b, b)$, where both agents select project b. In this case, each agent's utility is

$$u_i(\mathbf{s}^\star) = \frac{p_i(\{1, 2\}) \cdot r(b)}{|C_b(\mathbf{s}^\star)|} = \frac{K}{2}$$

Hence, the optimal social welfare is

$$\mathrm{OPT}(G) = \mathrm{SW}(\mathbf{s}^\star) = K$$

Combining the above, we obtain

$$\mathsf{PoA}_{\mathrm{NS}}(G) = \max_{\mathbf{s} \in \mathrm{NS}(G)} \frac{\mathrm{OPT}(G)}{\mathrm{SW}(\mathbf{s})} \geq \frac{K}{1} = K$$

Since K is arbitrary, the PoA grows without bound as $K \to \infty$. Consequently, the price of anarchy for the class $\mathscr{G}$ is unbounded. □

Proof of Theorem 3

Proof. Let $G \in \mathscr{G}$ be a hedonic project game with identical project rewards, which we normalize to $r(j) = 1$ for all $j \in M$. Assume that preferences are universal and concave monotonic decreasing, i.e., there exists a single function $f : \{1, \ldots, n\} \to \mathbb{R}_+$ such that $p_i(C) = f(|C|)$ for all agents i and coalitions $C \in \mathcal{N}_i$.

Let $\mathbf{s}$ be a Nash-stable outcome. By Lemma 1, $\mathbf{s}$ induces a coalition structure in which the sizes of all non-empty coalitions differ by at most one. In particular, $\mathbf{s}$ uses exactly $\min(n, m)$ projects, and all non-empty coalitions have size either

$$\left\lfloor \frac{n}{\min(n, m)} \right\rfloor \quad \text{or} \quad \left\lceil \frac{n}{\min(n, m)} \right\rceil$$

We now characterize the social welfare. Since rewards are identical and preferences are universal, social welfare can be written as

$$\mathrm{SW}(\mathbf{s}) = \sum_{j \in M(\mathbf{s})} \sum_{i \in C_j(\mathbf{s})} \frac{f(|C_j(\mathbf{s})|)}{|C_j(\mathbf{s})|} = \sum_{j \in M(\mathbf{s})} f(|C_j(\mathbf{s})|)$$

Thus, maximizing social welfare reduces to the problem of partitioning n agents into at most m coalitions to maximize $\sum_j f(|C_j|)$.

Because f is concave, the function $\sum_j f(|C_j|)$ is Schur-concave in the vector of coalition sizes. Hence, among all partitions of n agents into at most m coalitions, the social welfare is maximized by partitions whose coalition sizes are as balanced as possible, i.e., differ by at most one.

Therefore, every balanced coalition structure using $\min(n, m)$ projects attains maximum social welfare. Since $\mathbf{s}$ is Nash-stable and induces such a balanced structure, it follows that $\mathbf{s}$ is socially optimal (i.e., $\mathrm{SW}(\mathbf{s}) = \mathrm{OPT}(G)$).

Consequently, the best Nash-stable outcome achieves optimal social welfare, and $\mathsf{PoS}_{\mathrm{NS}}(G) = 1$. By Observation 2, Nash stability and joining stability coincide under monotonic decreasing preferences, implying

$$\mathsf{PoS}_{\mathrm{JS}}(G) = 1$$

□

Proof of Theorem 5

Proof. By definition, $\mathsf{JS}(G) \subseteq \mathsf{NS}(G)$, since every joining-stable outcome must in particular be Nash stable. It therefore suffices to show that $\mathsf{NS}(G) \subseteq \mathsf{JS}(G)$.

Let $\mathbf{s}$ be an arbitrary Nash-stable outcome. Consider any agent i and any unilateral deviation from project s_i to a project $j \neq s_i$. If $u_i(j, \mathbf{s}_{-i}) < u_i(\mathbf{s})$, then the deviation is unprofitable and hence irrelevant for joining stability. Suppose instead that agent i is indifferent, i.e.,

$$u_i(j, \mathbf{s}_{-i}) = u_i(\mathbf{s})$$

Let $\ell \in C_j(\mathbf{s})$ be any agent currently assigned to project j. Their utility before and after i joins is

$$u_\ell(\mathbf{s}) = p_\ell(C_j(\mathbf{s})) \cdot \frac{r(j)}{|C_j(\mathbf{s})|}$$

and

$$u_\ell(j, \mathbf{s}_{-i}) = p_\ell(C_j(\mathbf{s}) \cup \{i\}) \cdot \frac{r(j)}{|C_j(\mathbf{s})| + 1}$$

The joining-stability condition requires $u_\ell(\mathbf{s}) \geq u_\ell(j, \mathbf{s}_{-i})$, which is equivalent (since $r(j) > 0$) to

$$\frac{p_\ell(C_j(\mathbf{s}))}{|C_j(\mathbf{s})|} \geq \frac{p_\ell(C_j(\mathbf{s}) \cup \{i\})}{|C_j(\mathbf{s})| + 1}$$

This inequality holds by the assumption that preferences are per-capita non-increasing.

Since the above argument applies to every agent i, every project j, and every agent $\ell \in C_j(\mathbf{s})$, the outcome $\mathbf{s}$ satisfies the joining-stability condition. Hence, $\mathbf{s} \in \mathsf{JS}(G)$.

Therefore, $\mathsf{NS}(G) \subseteq \mathsf{JS}(G)$, and the two sets coincide. $\square$

Proof of Corollary 1

Proof. Under identical rewards $r(j) = r$ for all $j \in M$, the social welfare of any outcome $\mathbf{s}$ is:

$$\mathrm{SW}(\mathbf{s}) = \sum_{j \in M} \sum_{i \in C_j(\mathbf{s})} \hat{p}_i(C_j(\mathbf{s})) \cdot r = r \sum_{j \in M} \sum_{i \in C_j(\mathbf{s})} \hat{p}_i(C_j(\mathbf{s}))$$

Consider the outcome $\mathbf{s}^\star$ where all agents select the project $j^\star$ that maximizes $\sum_{i \in N} \hat{p}_i(N)$. Under per-capita non-decreasing preferences, for any agent i and any coalition $C \ni i$:

$$\hat{p}_i(N) \geq \hat{p}_i(C)$$

Thus, the grand coalition maximizes $\sum_{i \in N} \hat{p}_i(C)$ over all possible coalition structures. Hence $\mathbf{s}^\star$ is socially optimal.

By Theorem 4, $\mathbf{s}^\star$ is a Nash stable outcome, and every such outcome is also leaving stable under per-capita non-decreasing preferences. Since the optimal outcome is Nash stable, $\mathsf{PoS}_{\mathrm{NS}}(G) = \mathsf{PoS}_{\mathrm{LS}}(G) = 1$. $\square$

Proof of Corollary 2

Proof. Assume that preferences are per-capita strictly decreasing. Let $\mathbf{s}$ be an arbitrary outcome, and suppose that $\mathbf{s}$ contains a coalition $C_j(\mathbf{s})$ of size at least two for some project $j \in M$.

Fix any agent $i \in C_j(\mathbf{s})$, and consider the deviation in which agent i leaves project j and switches to an arbitrary project $j' \in M$ (possibly forming a singleton coalition). By definition, leaving stability requires that for every such deviation, and for every agent $\ell \in C_j(\mathbf{s})$, the inequality

$$u_\ell(\mathbf{s}) \geq u_\ell(j', \mathbf{s}_{-i})$$

must hold, regardless of whether the deviation is beneficial or harmful to the deviating agent i.

Let $\ell \in C_j(\mathbf{s}) \setminus \{i\}$ be any agent remaining in project j after the deviation. The utility of agent ℓ before and after the deviation is given by

$$u_\ell(\mathbf{s}) = p_\ell(C_j(\mathbf{s})) \cdot \frac{r(j)}{|C_j(\mathbf{s})|}$$

and

$$u_\ell(j', \mathbf{s}_{-i}) = p_\ell(C_j(\mathbf{s}) \setminus \{i\}) \cdot \frac{r(j)}{|C_j(\mathbf{s})| - 1}$$

respectively.

Since $r(j) > 0$, the leaving stability condition $u_\ell(\mathbf{s}) \geq u_\ell(j', \mathbf{s}_{-i})$ is equivalent to

$$\frac{p_\ell(C_j(\mathbf{s}))}{|C_j(\mathbf{s})|} \geq \frac{p_\ell(C_j(\mathbf{s}) \setminus \{i\})}{|C_j(\mathbf{s})| - 1}$$

or equivalently,

$$\hat{p}_\ell(C_j(\mathbf{s})) \geq \hat{p}_\ell(C_j(\mathbf{s}) \setminus \{i\})$$

However, since preferences are per-capita *strictly* decreasing and $C_j(\mathbf{s}) \setminus \{i\} \subset C_j(\mathbf{s})$, we have

$$\hat{p}_\ell(C_j(\mathbf{s})) < \hat{p}_\ell(C_j(\mathbf{s}) \setminus \{i\})$$

which contradicts the leaving stability condition. This argument holds for every choice of agent $i \in C_j(\mathbf{s})$ and every remaining agent $\ell \in C_j(\mathbf{s}) \setminus \{i\}$.

Therefore, no outcome containing a coalition of size at least two can satisfy leaving stability. It follows that in any leaving stable outcome, every coalition must be a singleton.

Finally, if $n > m$, then by the pigeonhole principle, at least one project must be selected by two or more agents in any outcome. Hence, no leaving stable outcome exists when $n > m$. $\square$